

# Introduction to Data Science and Engineering

## - Regression



Some materials courtesy of  
Rafael A. Irizarry, and are modified  
from the original version.

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Some slides retrofitted by Michael Wu



- Up to this point, we have focused mainly on single variables
- How to study the relationship between two or more variables?
- Let's try to understand regression by asking "is height hereditary?"
  - How well can we predict a child's height based on the parents' height?
  - Francis Galton developed the concepts of correlation and regression when he tried to answer the same question

The term "**regression**" in statistics was coined by **Sir Francis Galton** in the 19th century. It originally described a biological phenomenon where offspring tend to regress towards the mean height of their parents, rather than being as tall as the tallest parent or as short as the shortest. This concept was based on Galton's studies of the relationship between parents' and children's heights, leading to the statistical method we now know as regression analysis.



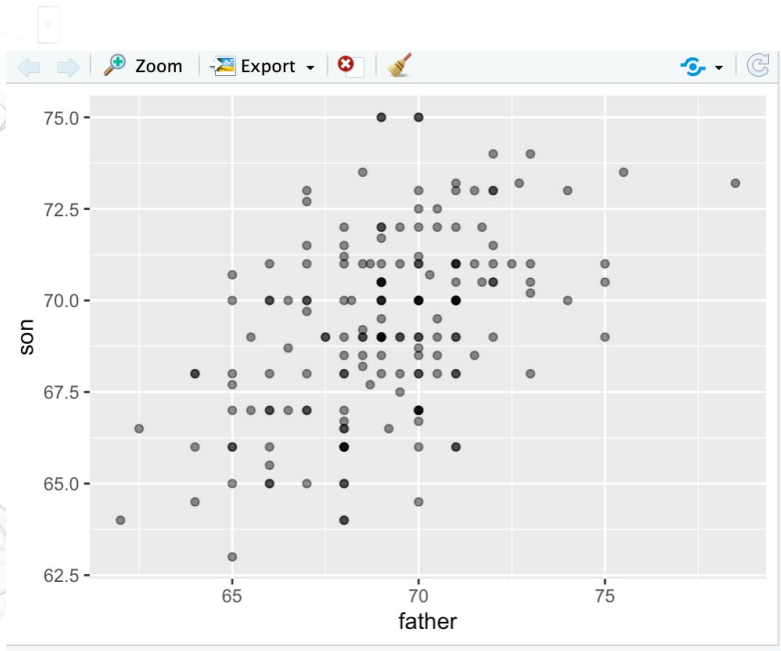
# Is height hereditary?

```
R 4.2.2 · ~/ ↵  
> library(tidyverse)  
> library(HistData)  
> data("GaltonFamilies")  
>  
> set.seed(42)  
> galton_heights <- GaltonFamilies |>  
+   filter(gender == "male") |>  
+   group_by(family) |>  
+   sample_n(1) |>  
+   ungroup() |>  
+   select(father, childHeight) |>  
+   rename(son = childHeight)
```

```
R 4.2.2 · ~/ ↵  
> galton_heights |>  
+   summarize(mean(father), sd(father), mean(son), sd(son))  
# A tibble: 1 × 4  
  `mean(father)` `sd(father)` `mean(son)` `sd(son)`  
    <dbl>         <dbl>         <dbl>         <dbl>  
1      69.1         2.55         69.3         2.50
```

We have access to Galton's family height data through the HistData package. This data contains heights on several dozen families: mothers, fathers, daughters, and sons. To imitate Galton's analysis, we will create a dataset with the heights of fathers and a randomly selected son of each family

Suppose we were asked to summarize the father and son data. Since both distributions are well approximated by the normal distribution, we could use the two averages and two standard deviations as summaries. However, this summary fails to describe an important characteristic of the data: the trend that the taller the father, the taller the son



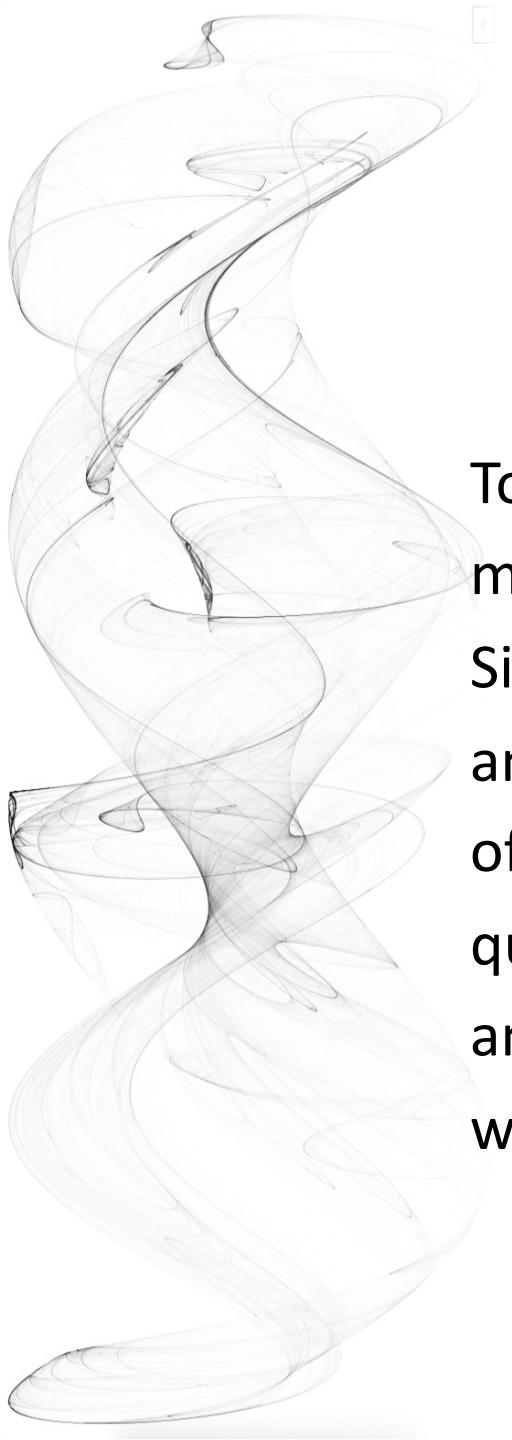
Correlation coefficient is an informative summary of how two variables move together and then see how this can be used to predict one variable using the other

$$\rho = \frac{1}{n} \sum_{i=1}^n \left( \frac{x_i - \mu_x}{\sigma_x} \right) \left( \frac{y_i - \mu_y}{\sigma_y} \right)$$

The correlation coefficient  $\rho$  is defined for a list of pairs  $(x_1, y_1), \dots, (x_n, y_n)$  **as the average of the product of the standardized values**.  $\mu_x, \mu_y$  are the average of  $x_1, \dots, x_n$ , and  $y_1, \dots, y_n$  respectively.  $\sigma_x, \sigma_y$  are the SDs

```
R 4.2.2 · ~/ ➡  
> rho <- mean(scale(x) * scale(y))  
rho <- cor(x,y)
```

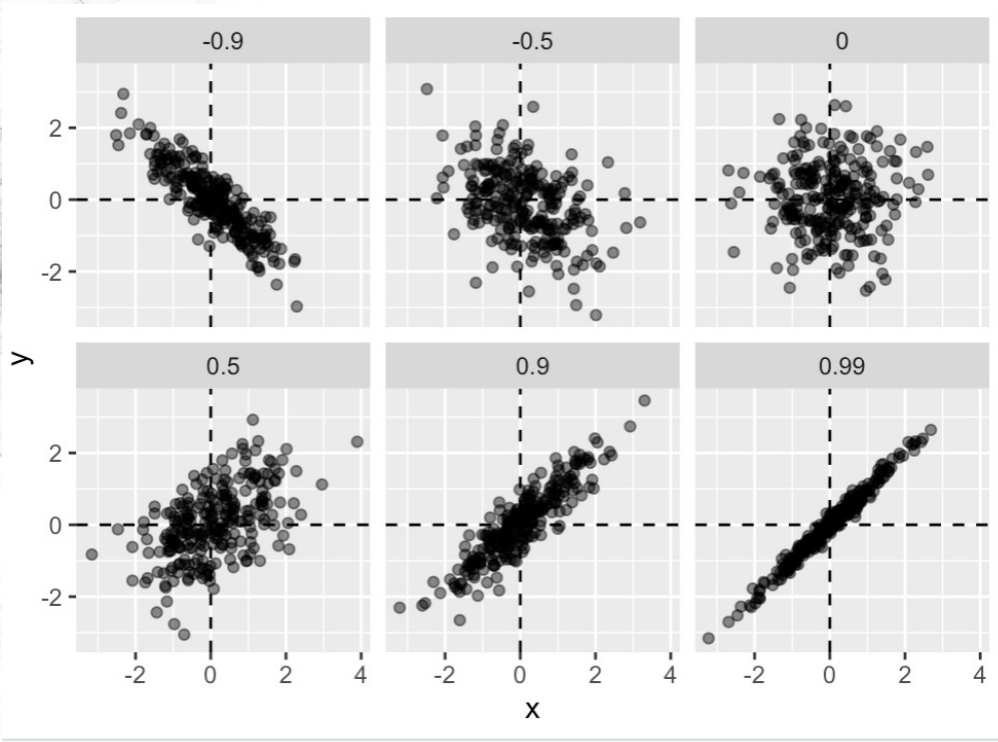
In R,  $\rho$  can be calculated in two ways


$$\rho = \frac{1}{n} \sum_{i=1}^n \left( \frac{x_i - \mu_x}{\sigma_x} \right) \left( \frac{y_i - \mu_y}{\sigma_y} \right)$$

To understand why this equation does in fact summarize how two variables move together, consider the  $i$ -th entry of  $x$  is  $\left( \frac{x_i - \mu_x}{\sigma_x} \right)$  SDs away from the average. Similarly, the  $y_i$  that is paired with  $x_i$ , is  $\left( \frac{y_i - \mu_y}{\sigma_y} \right)$  SDs away from the average. If  $x$  and  $y$  are unrelated, the product  $\left( \frac{x_i - \mu_x}{\sigma_x} \right) \left( \frac{y_i - \mu_y}{\sigma_y} \right)$  will be positive (+ $\times$ + and - $\times$ -) as often as negative (+ $\times$ - and - $\times$ +) and will average out to about 0. If instead the quantities vary together, then we are averaging mostly positive products (+ $\times$ + and - $\times$ -) and we get a positive correlation. If they vary in opposite directions, we get a negative correlation.

The correlation coefficient is always between -1 and 1. We can show this mathematically: consider that we can't have higher correlation than when we compare a list to itself (perfect correlation) and in this case the correlation is:

$$\rho = \frac{1}{n} \sum_{i=1}^n \left( \frac{x_i - \mu_x}{\sigma_x} \right)^2 = \frac{1}{\sigma_x^2} \frac{1}{n} \sum_{i=1}^n (x_i - \mu_x)^2 = \frac{1}{\sigma_x^2} \sigma_x^2 = 1$$



To see what data looks like for different values of  $\rho$ , here are six examples of pairs with correlations ranging from -0.9 to 0.99

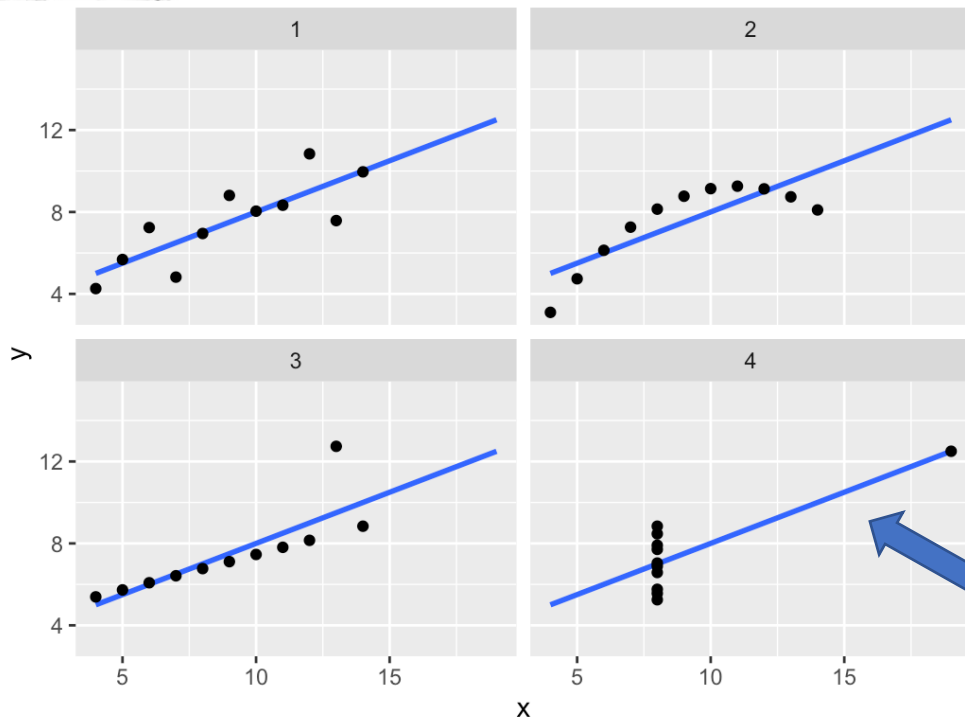


The correlation between father and son's heights is about 0.5

 R 4.2.2 · ~/ 

```
> galton_heights |> summarize(r = cor(father, son)) |> pull(r)  
[1] 0.4857975
```

Correlation is not always a good summary of the relationship between two variables. The following examples all show a correlation of 0.82



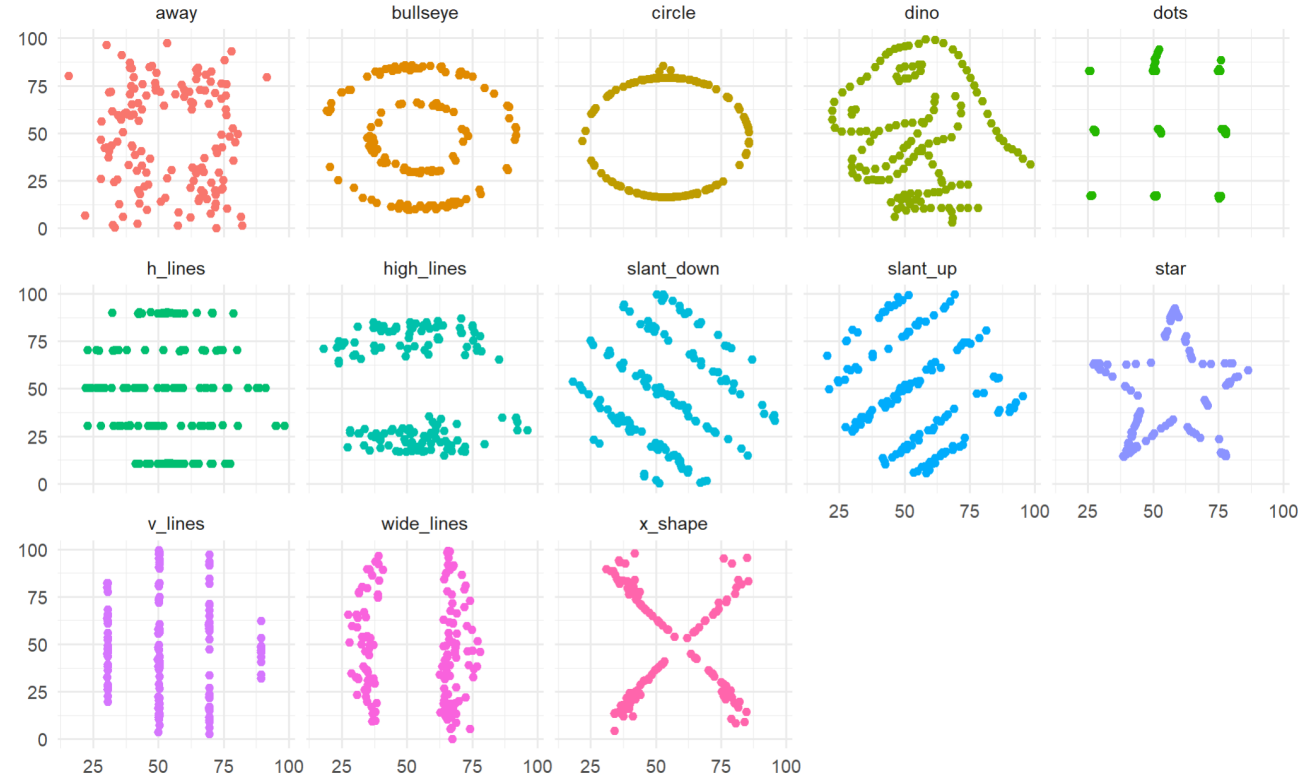
```
R 4.2.2 · ~/ 
> anscombe |> mutate(row = seq_len(n())) |>
+   gather(name, value, -row) |>
+   separate(name, c("axis", "group"), sep=1) |>
+   spread(axis, value) |> select(-row) |>
+   ggplot(aes(x,y)) +
+   facet_wrap(~group) +
+   geom_smooth(method="lm", fill=NA, fullrange=TRUE) +
+   geom_point()
```

This is a regression line

# Correlation coefficient

```
library(datasauRus)
datasaurus_dozen |>
  ggplot(aes(x, y, color = dataset)) +
  geom_point(show.legend = FALSE) +
  facet_wrap(~dataset, ncol = 5) +
  theme_minimal()
```

- Almost identical x-mean, x-sd, y-mean, y-sd
- Very similar  $\rho$





# Correlation coefficient

How to not misinterpret:

- **Visualize:** scatter plot is always a good helper
- Other correlation coefficients:
  - $\rho$  is also commonly known as Pearson correlation coefficient
  - Spearman's rank correlation coefficient: rank instead of z-score

```
cor(x, y, method="spearman")
```



# The regression line

If we are predicting a random variable  $Y$  knowing the value of another  $X = x$  using a regression line, then we predict that for every standard deviation,  $\sigma_X$ , that  $x$  increases above the average  $\mu_X$ ,  $Y$  increase  $\rho$  standard deviations  $\sigma_Y$  above the average  $\mu_Y$  with  $\rho$  the correlation between  $X$  and  $Y$ . The formula for the regression is therefore

$$\left( \frac{Y - \mu_Y}{\sigma_Y} \right) = \rho \left( \frac{x - \mu_X}{\sigma_X} \right)$$

We can rewrite it as

$$Y = \mu_Y + \rho \left( \frac{x - \mu_X}{\sigma_X} \right) \sigma_Y$$

If there is perfect correlation ( $\rho=1$ ), the regression line predicts an increase that is the same number of SDs. If there is 0 correlation, then we don't use  $x$  at all for the prediction and simply predict the average  $\mu_Y$ . For values between 0 and 1, the prediction is somewhere in between. If the correlation is negative, we predict a reduction instead of an increase



# The regression line

- For a father who is 180cm tall, how tall will his son be?
  - We can look at data from the male population that are around 180cm tall and summarize their sons' heights (modeling the conditional distribution).
  - Alternatively, we can build a regression model if we believe that there is a linear association between fathers' heights ( $x$ ) and sons' heights ( $y$ ) .

$$\left( \frac{Y - \mu_Y}{\sigma_Y} \right) = \rho \left( \frac{x - \mu_X}{\sigma_X} \right)$$

$$Y = \mu_Y + \rho \left( \frac{x - \mu_X}{\sigma_X} \right) \sigma_Y$$



# The regression line

- If we use  $y = b + mx$  to model the relationship between these two variables:

$$m = \rho \frac{\sigma_Y}{\sigma_X} \qquad b = \mu_Y - m\mu_X$$

- In the father and son case, SD of heights are the same, because  $\rho \in [-1, 1]$ ;  $y$  is always going to be closer to its mean than  $x$

## ANTHROPOLOGICAL MISCELLANEA.

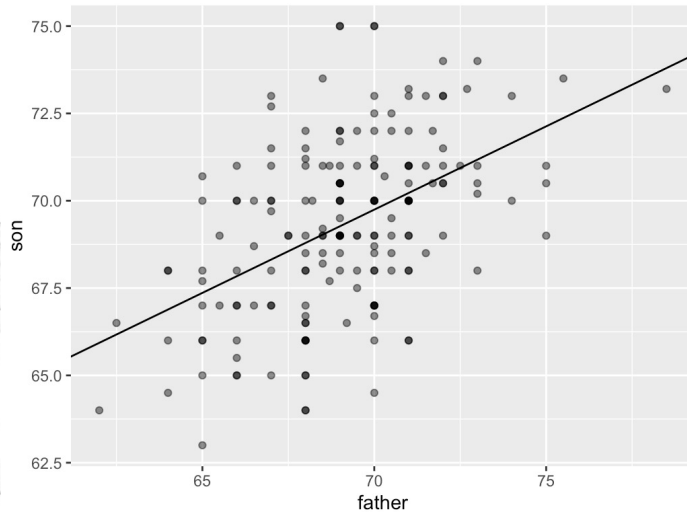
---

REGRESSION *towards* MEDIOCRITY *in* HEREDITARY STATURE.

By FRANCIS GALTON, F.R.S., &c.

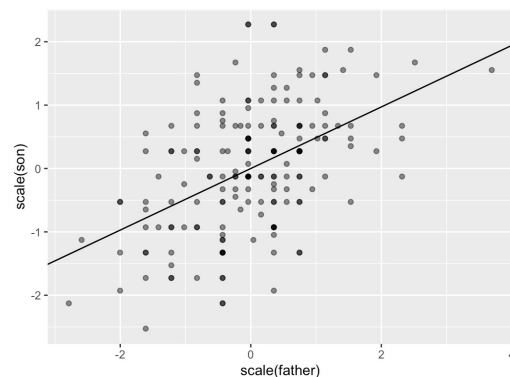
# The regression line

Here we add the regression line to the original data Galton's (son against father) data



```
R 4.2.2 · ~/ ↻  
> mu_x <- mean(galton_heights$father)  
> mu_y <- mean(galton_heights$son)  
> s_x <- sd(galton_heights$father)  
> s_y <- sd(galton_heights$son)  
> r <- cor(galton_heights$father, galton_heights$son)  
>  
> galton_heights |>  
+   ggplot(aes(father, son)) +  
+   geom_point(alpha = 0.5) +  
+   geom_abline(slope = r * s_y/s_x, intercept = mu_y - r * s_y/s_x * mu_x)
```

The regression formula implies that if we first standardize the variables, that is subtract the average and divide by the standard deviation, then the regression line has intercept 0 and slope equal to the correlation  $\rho$ . You can make same plot, but using standard units like this



```
R 4.2.2 · ~/ ↻  
> galton_heights |>  
+ ggplot(aes(scale(father), scale(son))) +  
+ geom_point(alpha = 0.5) +  
+ geom_abline(intercept = 0, slope = r)
```

# Regression improves prediction precision

Compare two methods to predict son's height using father's (72" specifically), which one is better

1. Round fathers' heights to closest inch, stratify, and then take the average
2. Compute the regression line and use it to predict.

We use a Monte Carlo simulation sampling  $N = 50$  families, as

## Method 1

```
R 4.2.2 · ~/ 
> rep <- 1000
> N <- 50
> set.seed(42)
> stratified_avg <- replicate(rep, {
+   dat <- sample_n(galton_heights, N)
+   dat |> filter(round(father) == 72) |>
+     summarize(avg = mean(son)) |>
+     pull(avg)
+ })
> mean(stratified_avg, na.rm = TRUE)
[1] 71.31454
> sd(stratified_avg, na.rm = TRUE)
[1] 0.724838
```

Similar mean, but **much lower SD**. The regression line is therefore much more stable than the stratified mean

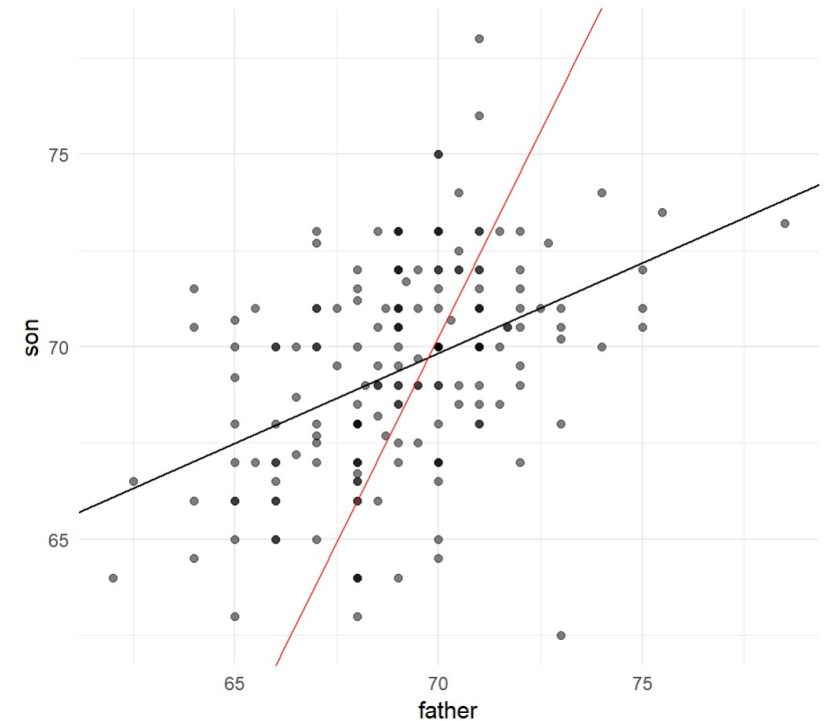
## Method 2

```
R 4.2.2 · ~/ 
> rep <- 1000
> N <- 50
> set.seed(42)
> regression_prediction <- replicate(rep, {
+   dat <- sample_n(galton_heights, N)
+   mu_x <- mean(dat$father)
+   mu_y <- mean(dat$son)
+   s_x <- sd(dat$father)
+   s_y <- sd(dat$son)
+   r <- cor(dat$father, dat$son)
+   mu_y + r*(72 - mu_x)/s_x*s_y
+ })
> mean(regression_prediction, na.rm = TRUE)
[1] 70.69053
> sd(regression_prediction, na.rm = TRUE)
[1] 0.3672109
```

# The regression line

- Regression coefficients are not commutative:
  - The regression line  $y = b + mx$  does NOT translate to  $x = \frac{y}{m} - \frac{b}{m}$

```
> model <- lm(son ~ father, data=galton_heights)
> coef(model)
(Intercept)      father
 37.0671243    0.4684554
> model <- lm(father ~ son, data=galton_heights)
> coef(model)
(Intercept)      son
40.5427394    0.4112533
```





Wait, how come that I am taller than my dad?

- The underlying formula is actually:

$$y = b + mx + \epsilon$$

- Here  $\epsilon$  (residue) is the part of  $y$  that is independent of (cannot be explained by  $x$ . (e.g., mother's height)

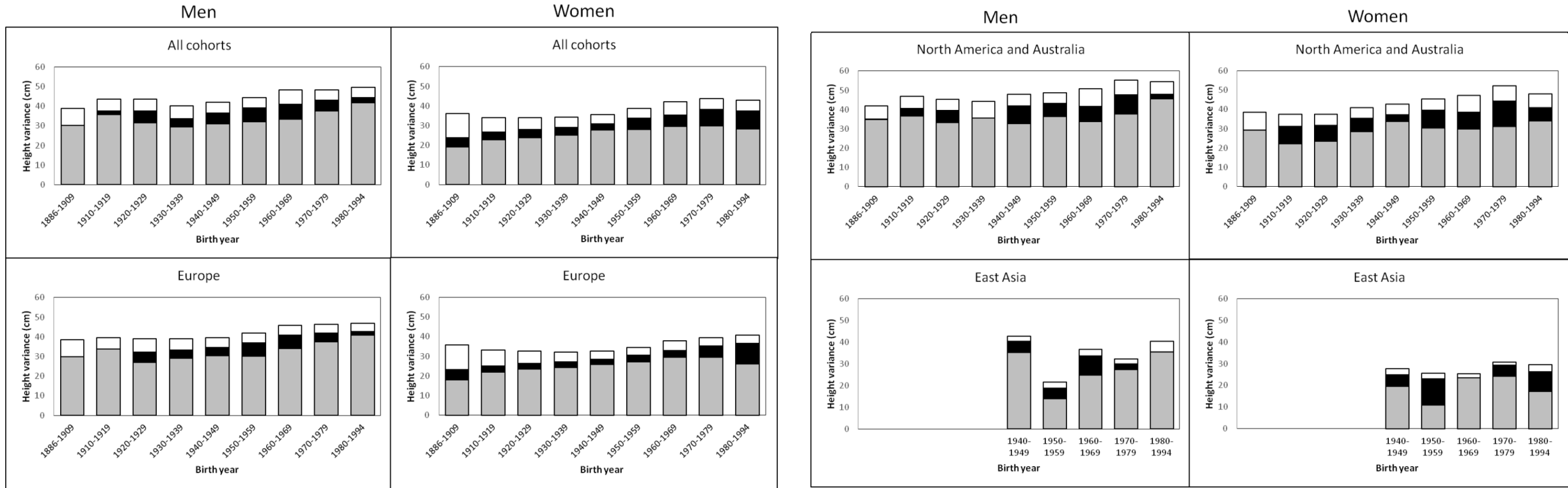
$$\text{Var}(y) = m^2 \text{Var}(x) + \text{Var}(\epsilon)$$

$$\text{Var}(y) = \left(\rho \frac{\sigma_Y}{\sigma_X}\right)^2 \text{Var}(x) + \text{Var}(\epsilon)$$

$$\text{Var}(y) = \rho^2 \text{Var}(y) + \text{Var}(\epsilon)$$

- Fathers' heights explained  $\rho^2$  (~25%) of the variance of sons' heights

# Is height hereditary?



- Additive genetic (grey), shared environmental (black) and unique environmental (white) variances of height across birth-year cohorts for the pooled data and by geographic-cultural region.
- (TL;DR) Yes, 70~80% is explained by genetic factors.



# Is height hereditary?

- Baseline: average of your mother's and father's heights, and:
  - Male: +6.5 cm
  - Female: -6.5cm
- What's your delta from the baseline?



Association is not causation

Association is not causation

Association is not causation



# Association is not causation

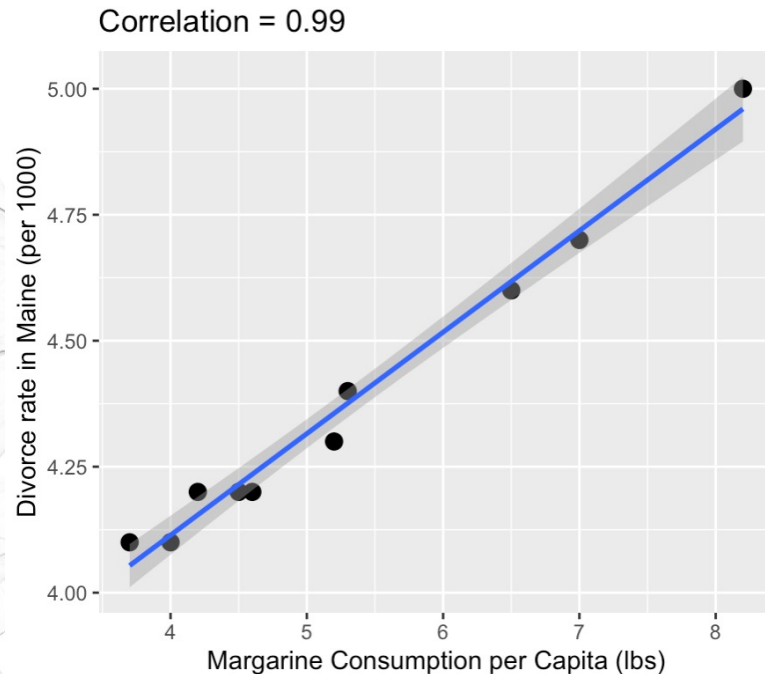
Association is not causation is perhaps the most important lesson one learns in a statistics class. Correlation is not causation is another way to say this. Throughout the Statistics part of the book, we have described tools useful for quantifying associations between variables. However, we must be careful not to over-interpret these associations

There are many reasons that a variable  $X$  can be correlated with a variable  $Y$  without having any direct effect on  $Y$ . Here we examine four common ways that can lead to misinterpreting data

## Examples

1. The more firemen are sent to a fire, the more damage is done.
2. Children who get tutored get worse grades than children who do not get tutored
3. In the early elementary school years, astrological sign is correlated with IQ, but this correlation weakens with age and disappears by adulthood

# Spurious correlation



The following comical example underscores that correlation is not causation. It shows a very strong correlation between divorce rates and margarine consumption

```
R 4.2.2 · ~/
> library(tidyverse)
> library(dslabs)
> the_title <- paste("Correlation =",
+                   round(with(divorce_margarine,
+                             cor(margarine_consumption_per_capita, divorce_rate_maine)),2))
> data(divorce_margarine)
> divorce_margarine |>
+   ggplot(aes(margarine_consumption_per_capita, divorce_rate_maine)) +
+   geom_point(cex=3) +
+   geom_smooth(method = "lm") +
+   ggtitle(the_title) +
+   xlab("Margarine Consumption per Capita (lbs)") +
+   ylab("Divorce rate in Maine (per 1000)")
```

More examples at <https://tylervigen.com/spurious-correlations>

We call these examples “data dredging”, “data fishing”, “data snooping”, or simply “cherry picking”

# Spurious correlation

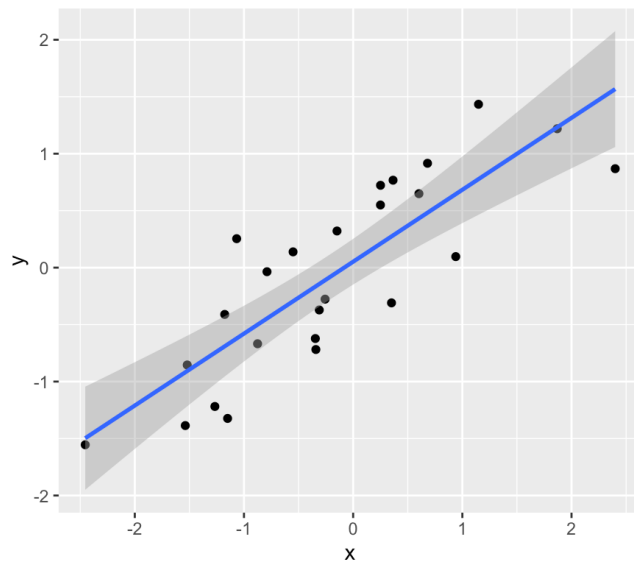
```
R 4.2.2 · ~/ 
> set.seed(42)
> N <- 25
> g <- 1000000
> sim_data <- data.frame(group = rep(1:g, each=N),
+                          x = rnorm(N * g),
+                          y = rnorm(N * g))
```

```
R 4.2.2 · ~/ 
> res <- sim_data |>
+   group_by(group) |>
+   summarize(r = cor(x, y)) |>
+   arrange(desc(r))
> res
# A tibble: 1,000,000 × 2
  group      r
  <int> <dbl>
1  117480 0.834
2  529225 0.810
3  548963 0.782
4  909788 0.781
5  550169 0.779
6  891349 0.766
7  422526 0.765
8  662756 0.760
9  630112 0.746
10 176141 0.744
```

A Monte Carlo simulation can be used to show how data dredging can result in finding high correlations among uncorrelated variables. We will save the results of our simulation into a dataframe. The first column denotes group. We created 1 million groups. For each one we generated a 25 pairs of X and Y each, stored in the second and third columns. Because we constructed the simulation, we know that X and Y are not correlated

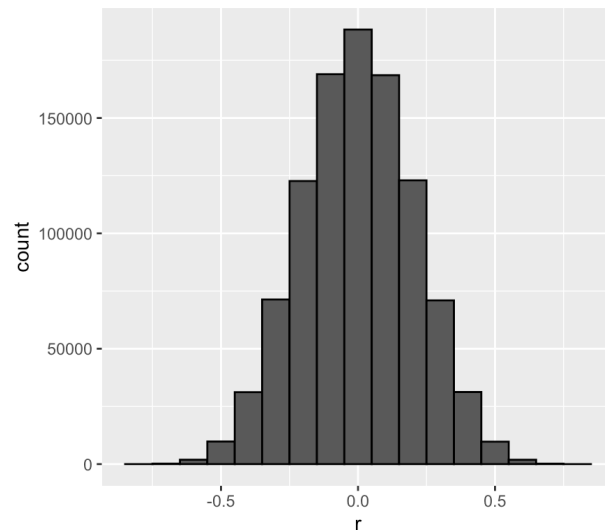
Next, we compute the correlation between X and Y for each group and look at the max. The max  $\rho$  is 0.834

# Spurious correlation



If you just plot the data from the group achieving the max correlation, it shows a convincing plot that  $X$  and  $Y$  are in fact correlated

```
R 4.2.2 · ~/ ➤  
> sim_data |> filter(group == res$group[which.max(res$r)]) |>  
+   ggplot(aes(x, y)) +  
+   geom_point() +  
+   geom_smooth(method = "lm")
```



Remember that the correlation summary is a random variable. Here is the distribution generated by the Monte Carlo simulation

```
R 4.2.2 · ~/ ➤  
> res |> ggplot(aes(x=r)) + geom_histogram(binwidth = 0.1, color = "black")
```

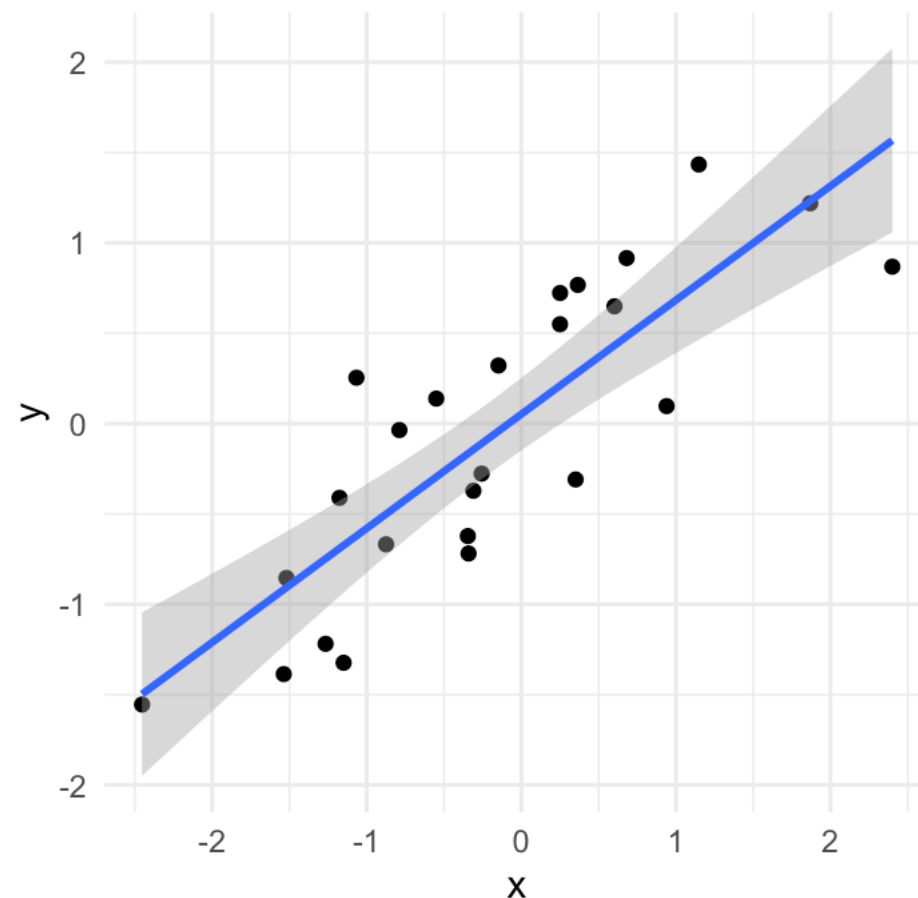
# Spurious correlation

- We can do statistical test to verify if null hypothesis (no correlation) can be rejected:

$$SD(\rho) = \sqrt{\frac{1 - \rho^2}{n - 2}}$$

$t = \rho / SD(\rho)$ , test against the t-distribution

```
> rho <- cor(subset$x, subset$y)
> t_stat <- rho / sqrt((1 - rho**2)/(N - 2))
> print(t_stat)
[1] 7.241302
> p_value <- 2 * pt(t_stat, df=N-1, lower.tail=FALSE)
> print(p_value)
[1] 1.756772e-07
```

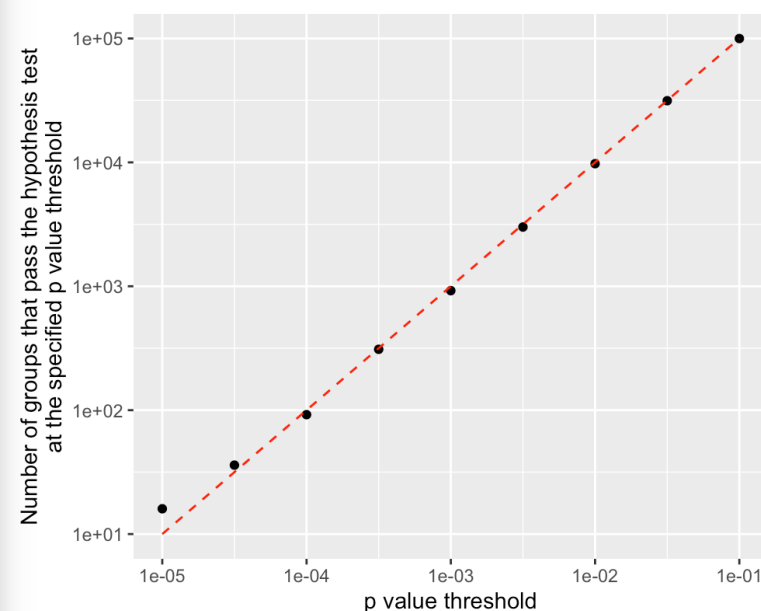


# What does p-value mean?

```
sim_data_with_p_value <- sim_data |>
  group_by(group) |>
  summarize(cor_pvalue = cor.test(x, y, method = "pearson")$p.value)

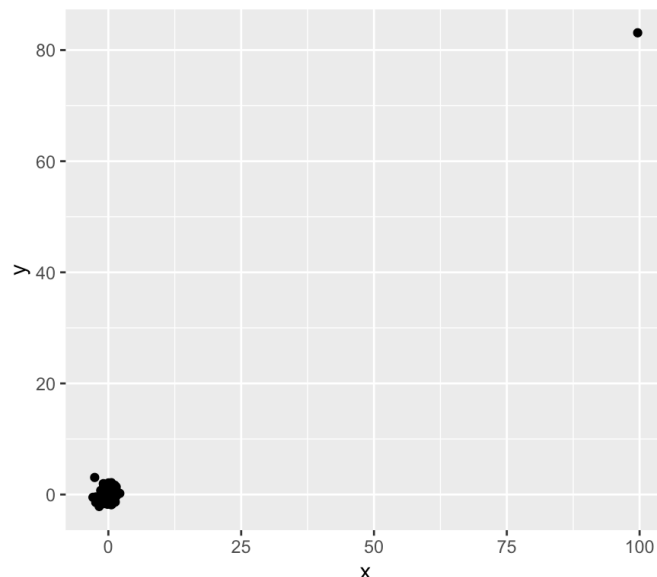
get_n_points_at_p_threshold <- function(p){
  sim_data_with_p_value |>
    filter(cor_pvalue <= p) |>
    pull(group) |>
    length()
}

df <- data.frame(p_thresholds = 10 ** seq(-5, -1, 0.5))
df |>
  mutate(n_points = sapply(p_thresholds, get_n_points_at_p_threshold)) |>
  ggplot(aes(p_thresholds, n_points)) +
    geom_point() +
    stat_function(fun = function(x) 1e6 * x, color = "red", linetype = "dashed") +
    scale_x_log10() +
    scale_y_log10() +
    xlab("p value threshold") +
    ylab("Number of groups passing the hypothesis test at the specified p-value thresholds")
```



- This applies to all other statistical tests.
- When we say p-value = 0.01, then on average (expectation) 1 out of 100 randomly generated datasets will pass the test.

# Outliers



```
R 4.2.2 · ~/   
> cor(x,y)  
[1] 0.9883282
```

```
R 4.2.2 · ~/   
> cor(x[-42], y[-42])  
[1] 0.02799956
```

Suppose we take measurements from two independent outcomes,  $X$  and  $Y$ , and we standardize the measurements. However, imagine we make a mistake and forget to standardize entry 42. We can simulate such data using

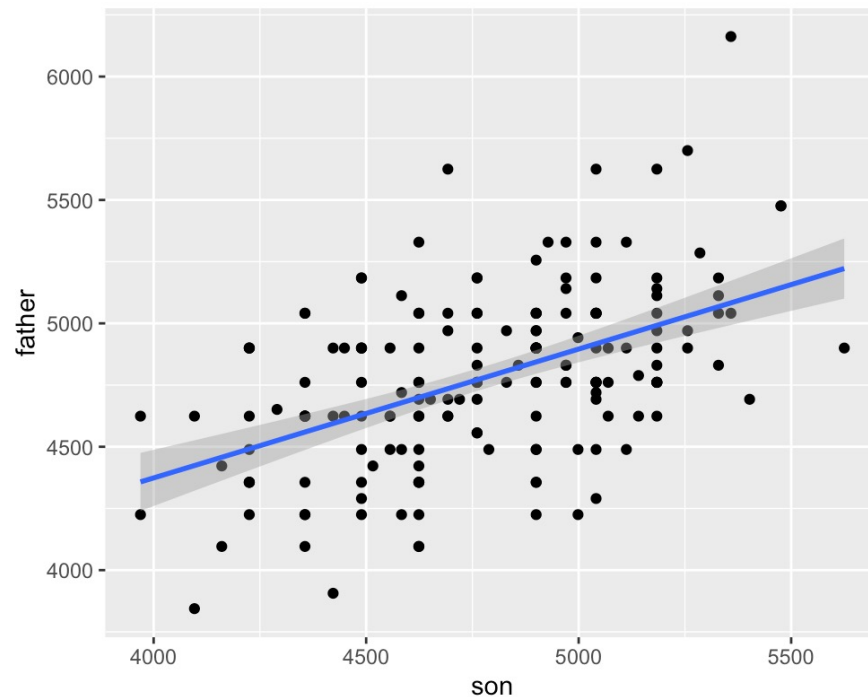
```
R 4.2.2 · ~/   
> set.seed(42)  
> x <- rnorm(100,100,1)  
> y <- rnorm(100,84,1)  
> x[-42] <- scale(x[-42])  
> y[-42] <- scale(y[-42])  
> qplot(x, y)
```

Not surprisingly, the correlation is very high

But this is driven by the one outlier. If we remove this outlier, the correlation is greatly reduced to almost 0, which is what it should be

# Reversing cause and effect

Another way association is confused with causation is when the cause and effect are reversed. We can easily construct an example of cause and effect reversal using the father and son height data, that is, to predict father's height using son's, and claim a tall son caused a tall father. Even though this correlation is true, we cannot say that it is because son is tall so that father is tall.

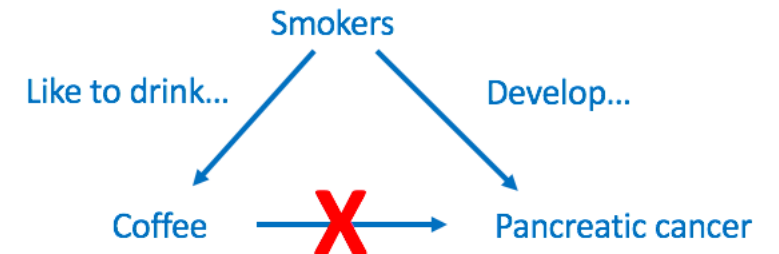
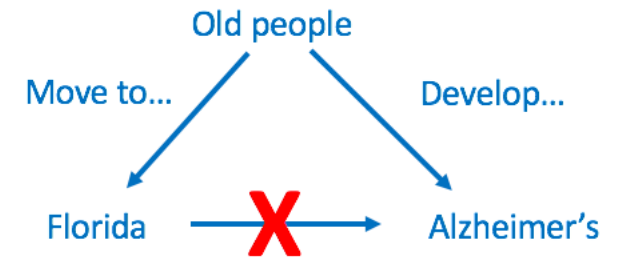


# Confounders

Confounders are perhaps the most common reason that leads to associations being misinterpreted

If  $X$  and  $Y$  are correlated, we call  $Z$  a confounder if changes in  $Z$  causes changes in both  $X$  and  $Y$ . In some cases, we can use linear models to account for confounders. However, this is not always the case

Incorrect interpretation due to confounders is ubiquitous in the lay press and they are often hard to detect. Here, we present a widely used example related to college admissions



# Confounders

Admission data from six U.C. Berkeley majors, from 1973, showed that more men were being admitted than women: 44% men were admitted compared to 30% women. PJ Bickel, EA Hammel, and JW O'Connell. Science (1975). We can load the data and a statistical test, which clearly rejects the hypothesis that gender and admission are independent

```
R 4.2.2 · ~/ 
> data(admissions)
> two_by_two <- admissions |> group_by(gender) |>
+   summarize(total_admitted = round(sum(admitted / 100 * applicants)),
+             not_admitted = sum(applicants) - sum(total_admitted)) |>
+   select(-gender)
>
> chisq.test(two_by_two)$p.value
[1] 1.055797e-21
```

But closer inspection shows a paradoxical result. Here are the percent admissions by major

```
R 4.2.2 · ~/ 
> admissions |> select(major, gender, admitted) |>
+   pivot_wider(names_from = "gender", values_from = "admitted") |>
+   mutate(women_minus_men = women - men)
# A tibble: 6 × 4
  major    men women women_minus_men
  <chr> <dbl> <dbl>         <dbl>
1 A      62    82             20
2 B      63    68              5
3 C      37    34             -3
4 D      33    35              2
5 E      28    24             -4
6 F       6     7              1
```

Four out of the six  
majors favor women

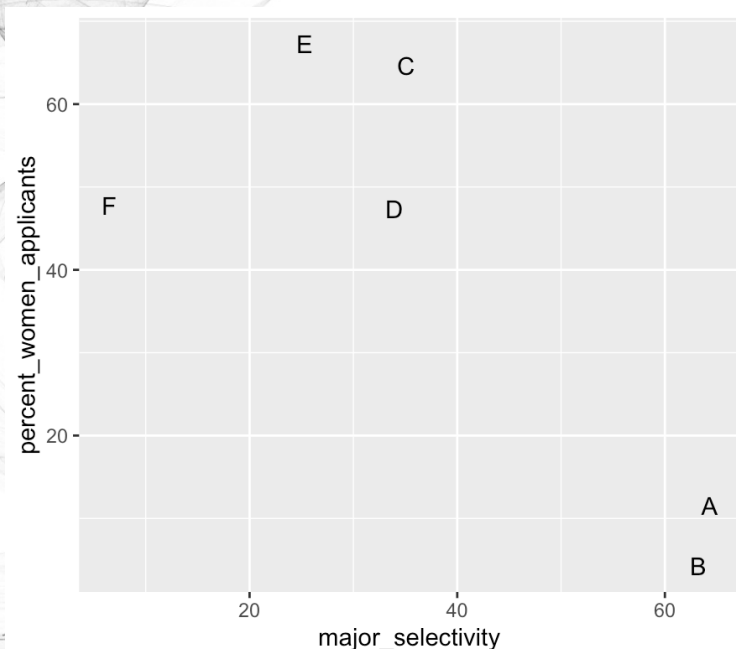
# Confounders

The paradox is that analyzing the totals suggests a dependence between admission and gender, but when the data is grouped by major, this dependence seems to disappear. What's going on?

**This actually can happen if an uncounted confounder is driving most of the variability**

A gender bias claim would be based on the fact that  $P(\text{admitted} | \text{male})$  is higher than  $P(\text{admitted} | \text{female})$ . However, major is an important confounder to consider. Clearly major is associated with admission rate, as the more selective a major, the lower  $P(\text{admitted})$ . But is major selectivity associated with gender?

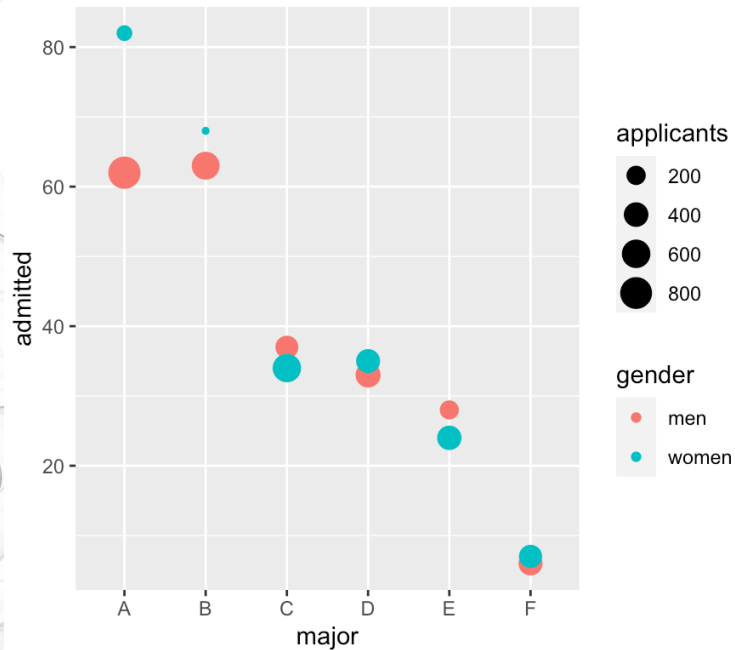
One way to see this is to plot the total percent admitted to a major versus the percent of women that made up the applicants



```
R 4.2.2 · ~/ 
> admissions |>
+ group_by(major) |>
+ summarize(major_selectivity = sum(admitted * applicants)/sum(applicants),
+           percent_women_applicants = sum(applicants * (gender=="women")) /
+           sum(applicants) * 100) |>
+ ggplot(aes(major_selectivity, percent_women_applicants, label = major)) +
+ geom_text()
```

There seems to be association. The plot suggests that women were much more likely to apply to the two “hard” majors: gender and major’s selectivity are confounded. Compare, for example, major B and major E. Major E is much harder to enter than major B and over 60% of applicants to major E were women, while less than 30% of the applicants of major B were women

# Confounders



In this plot, we can see that if we condition or stratify by major, and then look at differences, we control for the confounder and this effect goes away

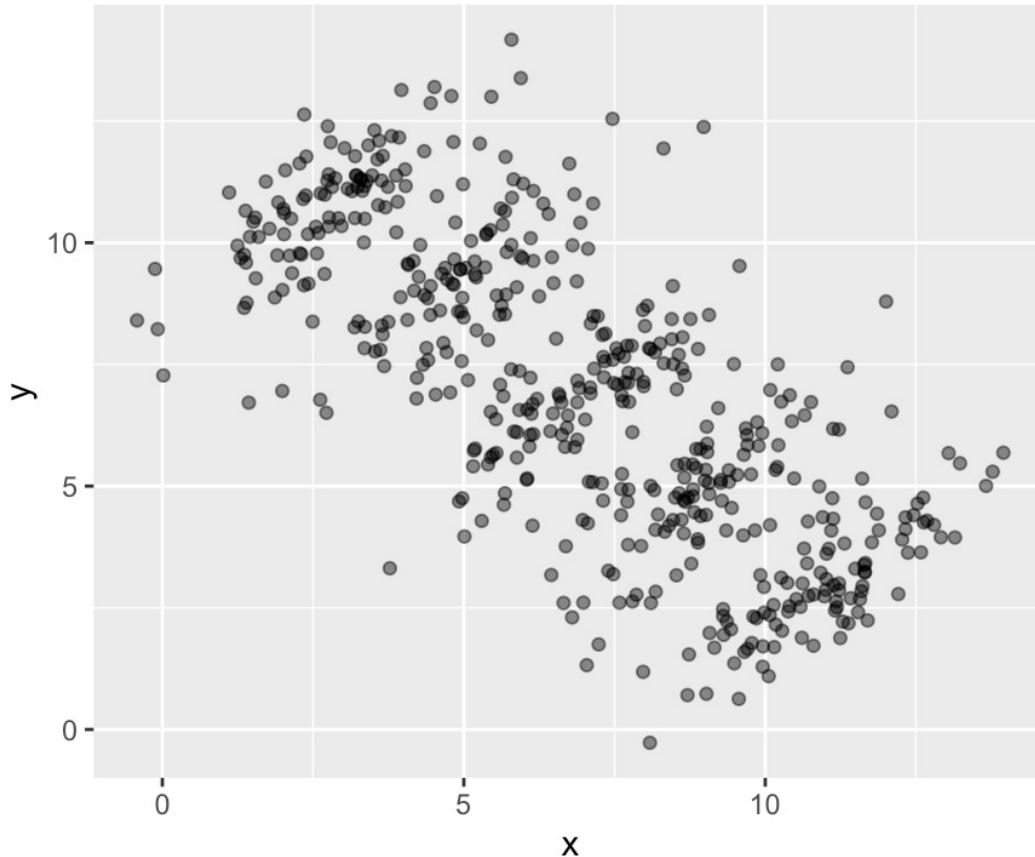
```
R 4.2.2 · ~/ ➡  
> admissions |>  
+ ggplot(aes(major, admitted, col = gender, size = applicants)) +  
+ geom_point()
```

If we average the difference by major, we find that the percent is actually 3.5% higher for women

```
R 4.2.2 · ~/ ➡  
> admissions |> group_by(gender) |> summarize(average = mean(admitted))  
# A tibble: 2 × 2  
  gender average  
  <chr>   <dbl>  
1 men     38.2  
2 women   41.7
```

# Simpson's paradox

Correlation = -0.72



The case we have just covered is an example of Simpson's paradox. It is called a paradox because we see the sign of the correlation flip when comparing the entire publication and specific strata. As an illustrative example, suppose you have three random variables  $X$ ,  $Y$ , and  $Z$  and that we observe realizations of these. On the left is a plot of simulated observations for  $X$  and  $Y$  along with the sample correlation. You can see that  $X$  and  $Y$  are negatively correlated

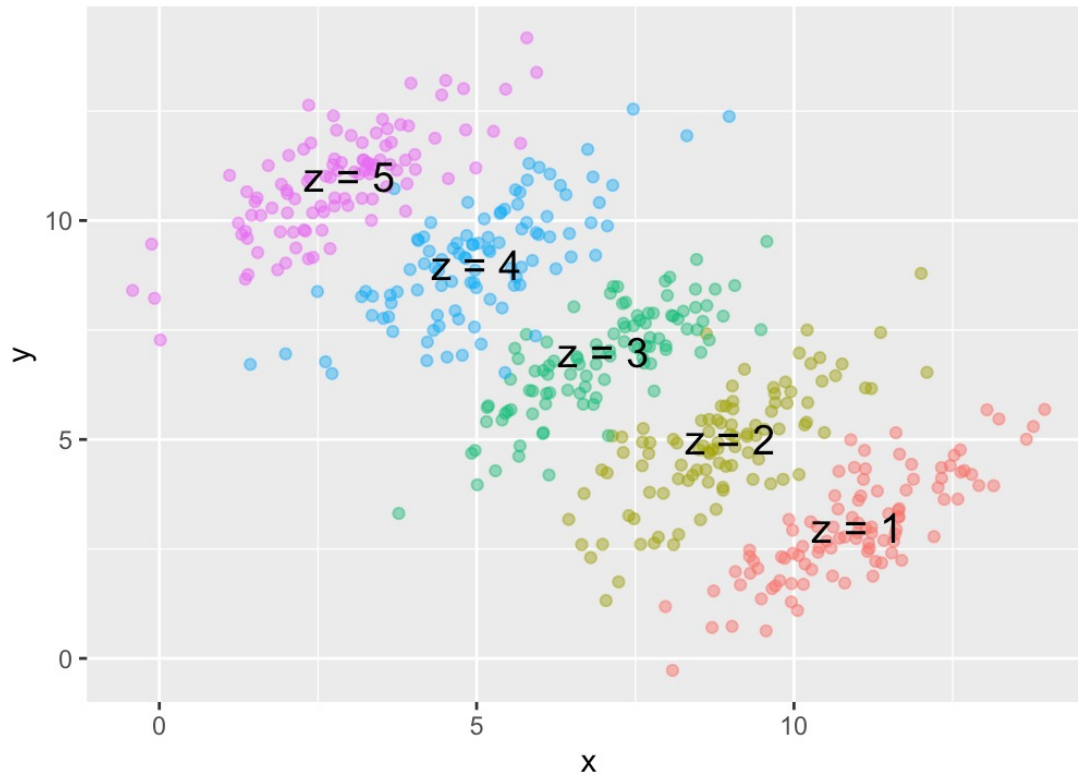
```
R 4.2.2 · ~/ 
> N <- 100
> Sigma <- matrix(c(1,0.75,0.75, 1), 2, 2)*1.5
> means <- list(c(x = 11, y = 3),
+               c(x = 9, y = 5),
+               c(x = 7, y = 7),
+               c(x = 5, y = 9),
+               c(x = 3, y = 11))
> dat <- lapply(means, function(mu){
+   res <- MASS::mvrnorm(N, mu, Sigma)
+   colnames(res) <- c("x", "y")
+   res
+ })
> dat <- do.call(rbind, dat) |>
+   as_tibble() |>
+   mutate(z = as.character(rep(seq_along(means), each = N)))
> dat |> ggplot(aes(x, y)) + geom_point(alpha = 0.5) +
+   ggtitle(paste("Correlation = ", round(cor(dat$x, dat$y), 2)))
```

# Simpson's paradox

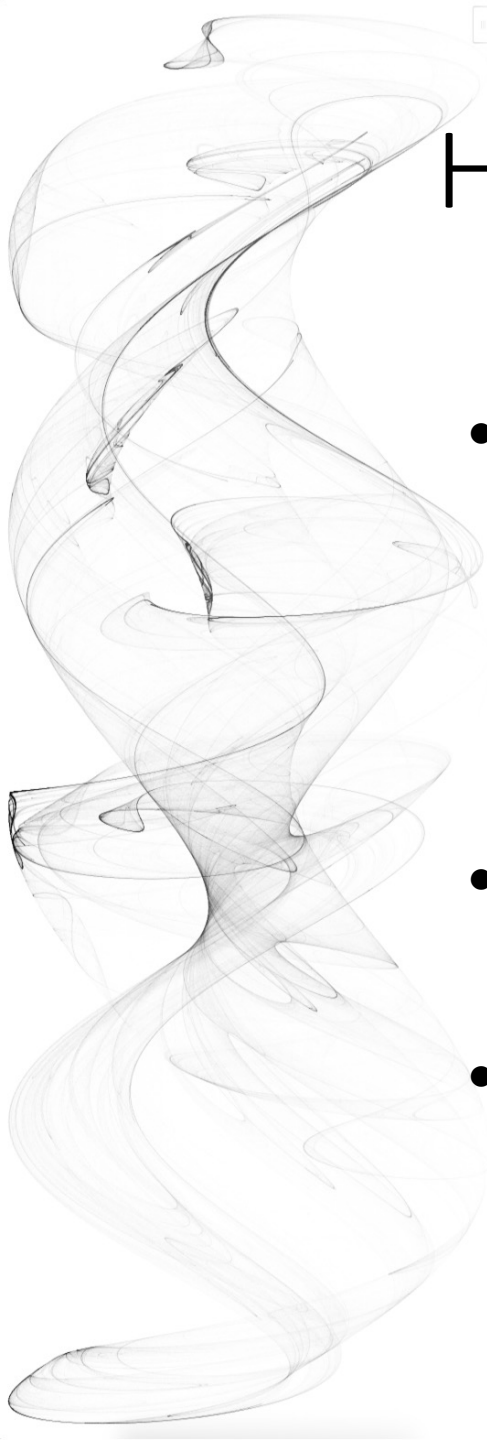
However, once we stratify by  $Z$  (shown in different colors) another pattern emerges

It is really  $Z$  that is negatively correlated with  $X$ . If we stratify by  $Z$ , the  $X$  and  $Y$  are actually positively correlated as seen in the plot above

Correlations = 0.81 0.74 0.79 0.69 0.76



```
R 4.2.2 · ~/ 
> means <- do.call(rbind, means) |>
+   as_tibble() |>
+   mutate(z = as.character(seq_along(means)))
>
> corrs <- dat |> group_by(z) |> summarize(cor = cor(x, y)) |> pull(cor)
> dat |> ggplot(aes(x, y, color = z)) +
+   geom_point(show.legend = FALSE, alpha = 0.5) +
+   ggtitle(paste("Correlations =", paste(signif(corrs,2), collapse="
+   "))) +
+   annotate("text", x = means$x, y = means$y, label = paste("z =", means
+   $z), cex = 5)
```

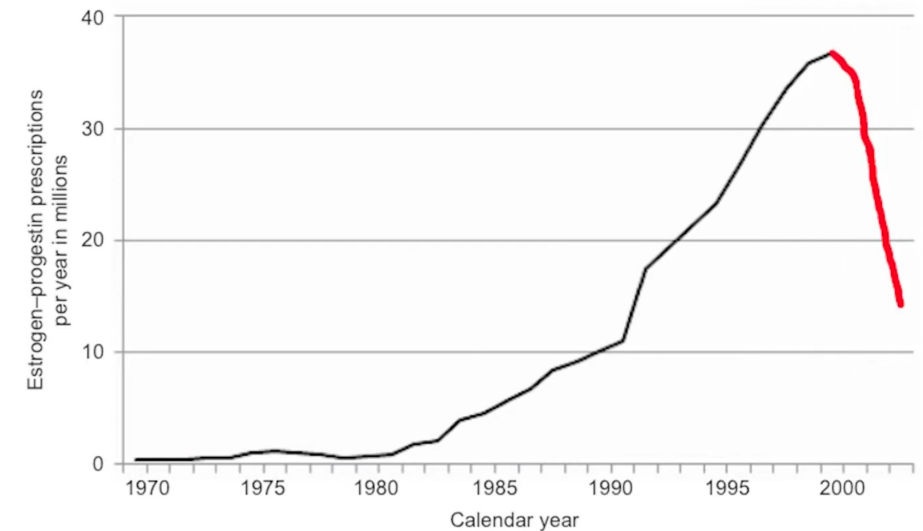


# How do we derive true causal relationship

- Randomized Controlled Trials (RCTs): the gold standard
  - We don't control for confounders; we just use random sampling to make sure that confounder effects are the same between groups
  - The treatment ( $X$ ) is applied after random assignment of groups, so it is strictly independent of all possible confounding factors ( $Z$ ).
- Twins, difference-in-differences
  - Minimize chance of potential confounders
- Causal inference methods: propensity score matching, graphical modeling, etc.

# Observational study vs RCT

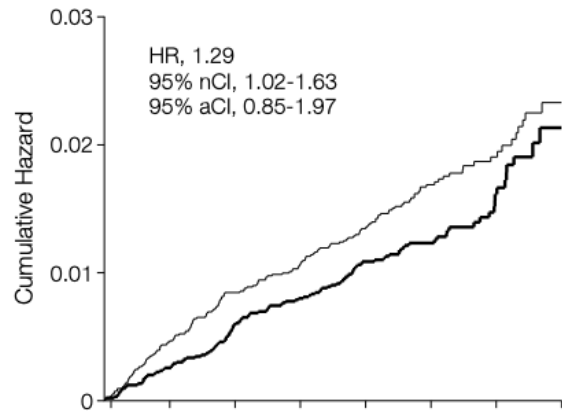
- In 1980s, large observational studies show that postmenopausal women on hormone replacement therapies had lower CVD risk.
- However, this was later found to be confounded by other factors; women on therapies are wealthier and more health-conscious.
- Randomized controlled trials later in 1990s and 2000s show that such therapies might increase CVD risks.



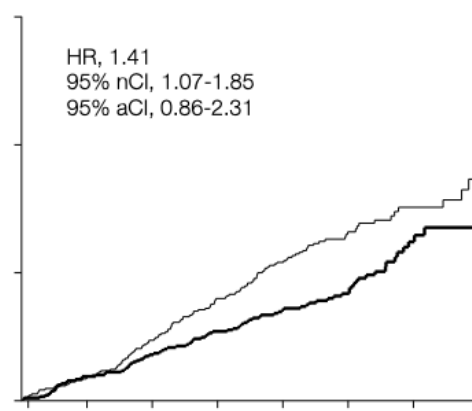
To learn more: <https://www.youtube.com/watch?v=MVYWqWu2Za4>

— Estrogen + Progestin — Placebo

Coronary Heart Disease



Stroke



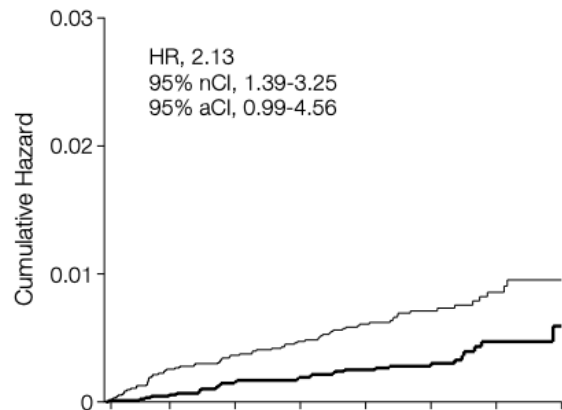
- RCT shows higher risk of CHD, Stroke in patients taking estrogen + progestin

No. at Risk

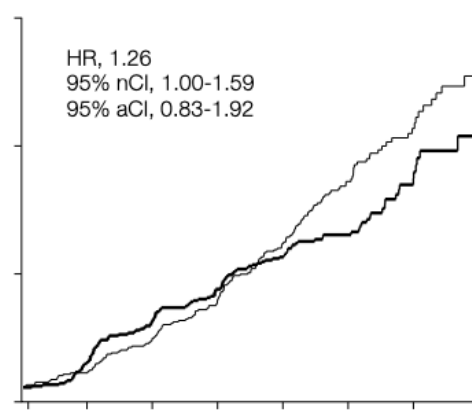
| Time (y)             | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7   |
|----------------------|------|------|------|------|------|------|------|-----|
| Estrogen + Progestin | 8506 | 8353 | 8248 | 8133 | 7004 | 4251 | 2085 | 814 |
| Placebo              | 8102 | 7999 | 7899 | 7789 | 6639 | 3948 | 1756 | 523 |

| Time (y)             | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7   |
|----------------------|------|------|------|------|------|------|------|-----|
| Estrogen + Progestin | 8506 | 8375 | 8277 | 8155 | 7032 | 4272 | 2088 | 814 |
| Placebo              | 8102 | 8005 | 7912 | 7804 | 6659 | 3960 | 1760 | 524 |

Pulmonary Embolism



Invasive Breast Cancer

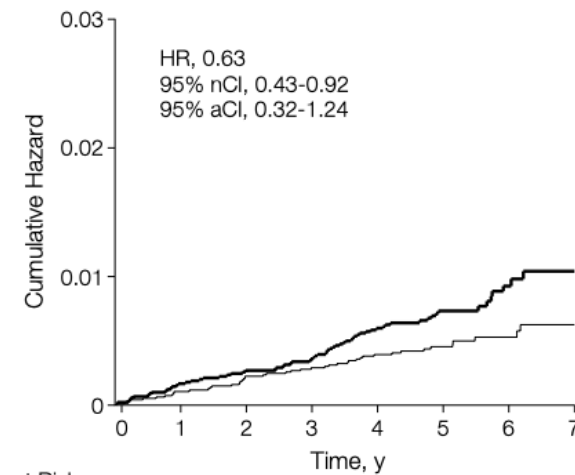


No. at Risk

| Time (y)             | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7   |
|----------------------|------|------|------|------|------|------|------|-----|
| Estrogen + Progestin | 8506 | 8364 | 8280 | 8174 | 7054 | 4295 | 2108 | 820 |
| Placebo              | 8102 | 8013 | 7924 | 7825 | 6679 | 3973 | 1770 | 526 |

| Time (y)             | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7   |
|----------------------|------|------|------|------|------|------|------|-----|
| Estrogen + Progestin | 8506 | 8378 | 8277 | 8150 | 7000 | 4234 | 2064 | 801 |
| Placebo              | 8102 | 8001 | 7891 | 7772 | 6619 | 3922 | 1740 | 523 |

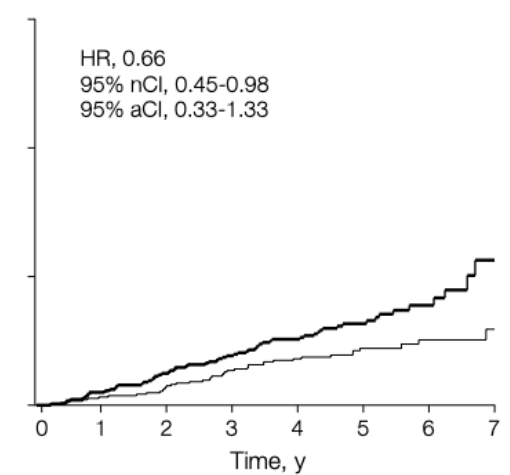
Colorectal Cancer



No. at Risk

| Time (y)             | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7   |
|----------------------|------|------|------|------|------|------|------|-----|
| Estrogen + Progestin | 8506 | 8379 | 8297 | 8194 | 7073 | 4305 | 2111 | 825 |
| Placebo              | 8102 | 8003 | 7916 | 7814 | 6660 | 3958 | 1756 | 522 |

Hip Fracture



| Time (y)             | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7   |
|----------------------|------|------|------|------|------|------|------|-----|
| Estrogen + Progestin | 8506 | 8382 | 8299 | 8190 | 7073 | 4305 | 2116 | 826 |
| Placebo              | 8102 | 8009 | 7915 | 7807 | 6659 | 3958 | 1763 | 525 |

Rossouw, Jacques E., et al. "Risks and benefits of estrogen plus progestin in healthy postmenopausal women: principal results From the Women's Health Initiative randomized controlled trial." *Jama* 288.3 (2002): 321-333.